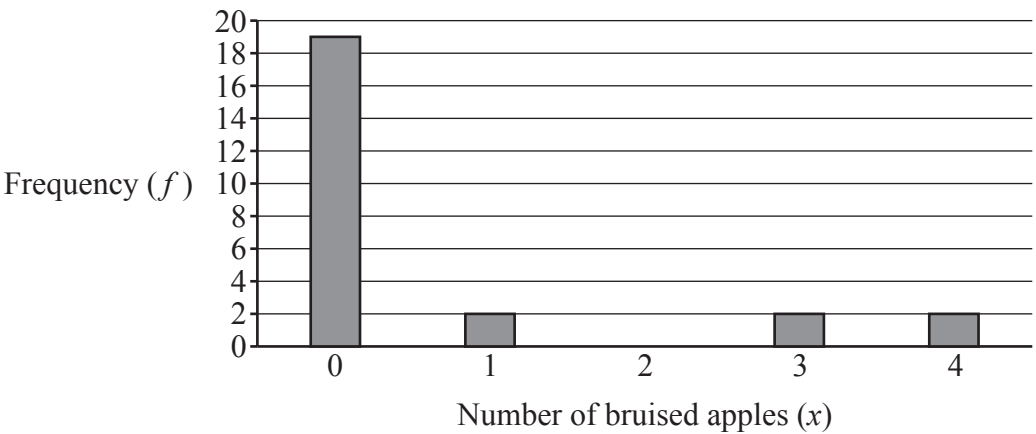
11 Apples are sold in packets of 4. Ling and Sam are investigating the frequency, f, of the number of bruised apples in a packet, x. They each collect a random sample of packets, note the number of bruised apples in each packet and draw a diagram to represent their data.

Ling's results are shown in **Table 11.1** and **Fig. 11.1**.

Table 11.1

x	0	1	2	3	4
f	19	2	0	2	2

Fig. 11.1

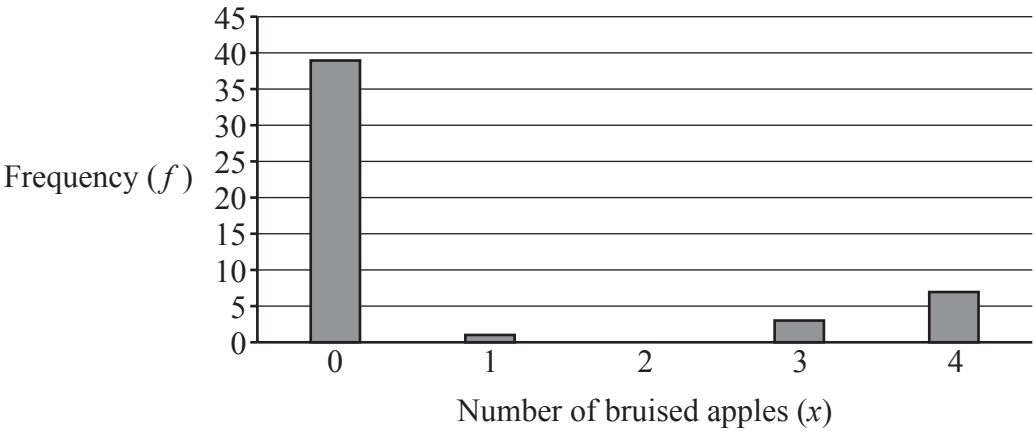


Sam's results are shown in **Table 11.2** and **Fig. 11.2**.

Table 11.2

x	0	1	2	3	4
f	39	1	0	3	7

Fig. 11.2



(a) Explain why Sam's diagram is likely to be a better representation of the true distribution of the number of bruised apples in a packet than Ling's diagram. **[1]**

(b) Calculate the mean number of bruised apples per packet for Sam's data. **[1]**

Sam thinks that the distribution of bruised apples may be modelled by a binomial distribution.

(c) Use your answer to part **(b)** to calculate the value of p, the probability that an apple selected at random is bruised, for Sam's model. **[1]**

(d) Calculate the theoretical frequency distribution of the number of bruised apples per packet for Sam's model, giving your answers correct to **2** decimal places. **[3]**

(e) Comment on whether Sam's model appears to be a good fit for the data. **[1]**